Fixed-G Inference with Climate-Based Clusters\ Advanced Econometrics II: Non-standard Inference and Simulation*

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Introduction

Bester et al. (2011), hereafter BCH, study inference with dependent data using cluster covariance estimators. Their contribution is a fixed-G theory for settings where observations are divided into a few large groups. The covariance matrix is estimated from group score sums, and critical values reflect that only a small number of groups are available. A central result is that, if G is fixed and group size grows, the cluster covariance estimator remains random. For a scalar coefficient, the BCH statistic is therefore compared with t_{G-1}, not $N(0,1)$. The practical requirement is not simply many clusters. It is large, well-shaped groups whose score sums are approximately independent.

This report uses that result to ask one empirical question with two parts: does the number of groups change inference in a spatial climate-conflict application, and how can groups be chosen when small spatial units are correlated but the true dependence structure is unknown? The setting comes from Burke et al. (2024), who study whether wealth weakens the link between weather shocks and civil conflict in Africa. Their replication package constructs a spatially disaggregated panel from Berkeley Earth temperature data, UCDP georeferenced conflict events, local wealth estimates, and national GDP per capita. We use the public weather-conflict component of this package. The broader climate-conflict literature summarized by Hsiang et al. (2013) remains useful context, but Burke et al. (2024) provide the direct empirical motivation.

The report proceeds in four steps. First, a simulation reproduces the fixed-G mechanism: changing G changes both variance estimation and the reference distribution. Second, candidate regions are constructed from temperature-anomaly co-movement using HiClimR (Badr et al., 2015). The groups are based on treatment dynamics, not conflict outcomes. Third, a fixed-effects climate-conflict model is estimated once, and BCH-style inference is recomputed for $G = 4, 8, 12, 16, 20$. Fourth, score independence is compared across data-driven climate groups, supranational administrative macroregions, and country clusters.

Section 2 explains why fixed-G inference differs from normal cluster inference. Section 3 applies the question to the Burke et al. weather-conflict data: it describes the data, constructs the climate groups, estimates the model, and compares score diagnostics against administrative groupings. Section 4 states the resulting interpretation and its limits.

Monte Carlo Simulation for Fixed-G Cluster Inference

The notation below is deliberately stylized to isolate the inferential mechanism in BCH. It defines a spatial lattice in which both the regressor and the error are locally dependent, estimates OLS under a true null, and asks what happens when inference is based on a fixed number G of large spatial groups. The objects of interest are the group-level covariance estimator and the studentized coefficient.

Let $\mathcal{S}_K = \{1,\dots,K\}^2 \subset \mathbb{Z}^2$, $N_K = K^2$, and

$$d_\infty(s,t) = \max\{|s_1 - t_1|,\ |s_2 - t_2|\}.$$

On a two-cell padding of $\mathcal{S}_K$, let u_r and v_r be independent Gaussian white-noise fields. For $\gamma \in \{0, 0.3, 0.6\}$, define the spatial moving-average fields

$$x_\gamma(s) = \sum_{d_\infty(h,0)\le 2} \gamma^{d_\infty(h,0)} v_{s+h}, \qquad \varepsilon_\gamma(s) = \sum_{d_\infty(h,0)\le 2} \gamma^{d_\infty(h,0)} u_{s+h}.$$

The outcome is

$$y_\gamma(s) = \beta_0 x_\gamma(s) + \varepsilon_\gamma(s), \qquad \beta_0 = 1,$$

so $H_0 : \beta = \beta_0$ is true in every replication. Let X_K have row $(1, x_\gamma(s))$, and write the OLS estimator as

$$\widehat{\theta}_K = (X_K' X_K)^{-1} X_K' Y_K, \qquad \widehat{\theta}_K = (\widehat{\alpha}_K, \widehat{\beta}_K)'.$$

Partition $\mathcal{S}_K$ into G equal contiguous blocks. With $X_{g,K}$ and $\widehat{e}_{g,K}$ denoting observations and residuals in group g, the cluster covariance estimator is

$$\widehat{V}_K^{CCE} = \frac{1}{N_K} \sum_{g=1}^{G} X_{g,K}' \widehat{e}_{g,K} \widehat{e}_{g,K}' X_{g,K}.$$

*https://github.com/Thomas-Bombarde/Adv.Metrics-Assignment-2026

Let $\widehat{Q}_K = N_K^{-1} X_K' X_K$, and $a = (0,1)'$. The raw CCE statistic is

$$T_K^{CCE} = \frac{\sqrt{N_K}(a'\widehat{\theta}_K - \beta_0)}{\left[a'\widehat{Q}_K^{-1}\widehat{V}_K^{CCE}\widehat{Q}_K^{-1}a\right]^{1/2}},$$

whereas the BCH statistic rescales it as

$$T_K^{BCH} = \sqrt{\frac{G-1}{G}}\, T_K^{CCE} \quad \text{and is compared with } t_{G-1}.$$

The interpretation follows the main BCH result. As group size grows, each group score becomes approximately normal, and well-separated groups become approximately independent. But if G itself is fixed, the covariance matrix is still estimated from only G group scores. The cluster covariance estimator therefore remains random rather than converging to a deterministic variance. The BCH rescaling and the t_{G-1} reference distribution account for that remaining uncertainty.

Simulation Evidence BCH use an asymptotic sequence with fixed G and $L_K = N_K/G \to \infty$. The following simulations keep the design deliberately simple so that the only issue is inference. Within each group, the group score satisfies a central limit theorem. Across well-shaped groups, scores become approximately independent. Since only G scores estimate the variance, $\widehat{V}_K^{CCE}$ does not converge to a constant. It has a random limit.

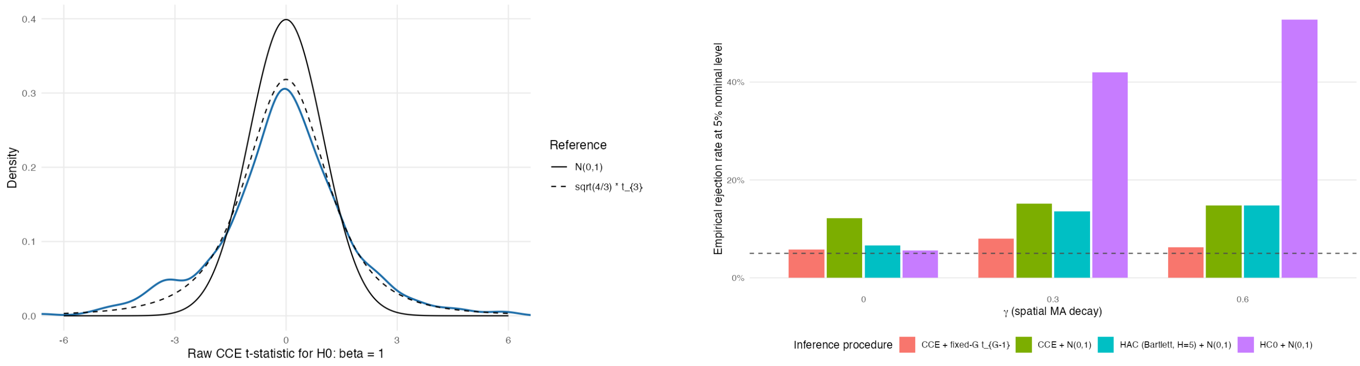

Figure 1: Fixed-G Monte Carlo evidence. Left: raw CCE t-statistic for $K = 36$, $\gamma = 0.6$, $G = 4$; the BCH reference is wider than $N(0,1)$. Right: rejection rates for nominal 5 percent tests; the BCH correction is closer to the target size.

Figure 1 gives the operational implication. Normal critical values treat the denominator as precise. BCH use t_{G-1} critical values because the variance is estimated from only G group scores. This matters in spatial panels. Larger groups can improve cross-group independence, but they also reduce G. Fixed-G inference targets this trade-off.

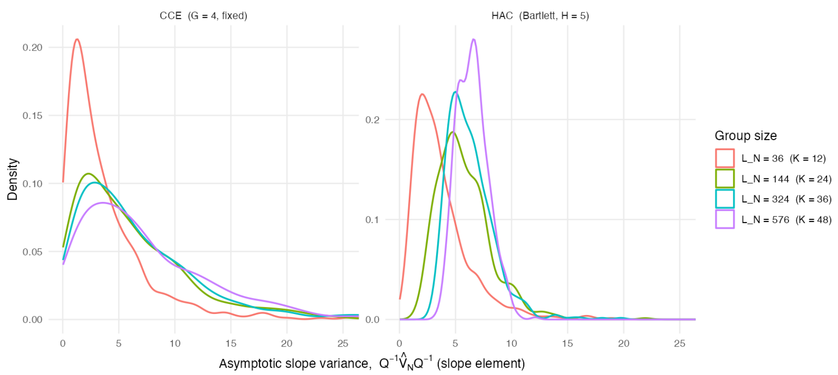

Figure 2: Sampling distribution of the asymptotic-scale variance object. With fixed $G = 4$, the CCE variance estimate remains dispersed as group size grows; the issue is variance-estimator randomness, not only point-estimator convergence.

Figure 2 should not be read as saying that the CCE is better because its slope-variance distribution is lower. Precision is valuable only if the covariance estimator is valid for the dependence structure. The figure is useful for a different reason. It separates two variance questions. The actual sampling variance of $\widehat{\beta}$ shrinks with N. The plotted object is instead the asymptotic-scale variance used to studentize $\sqrt{N}(\widehat{\beta} - \beta_0)$. Under fixed G, the CCE estimate of this object remains random because it is built from only G group scores. A lower realization of this object means a smaller standard error in that replication; it does not by itself mean that the grouping has uncovered more independent information.

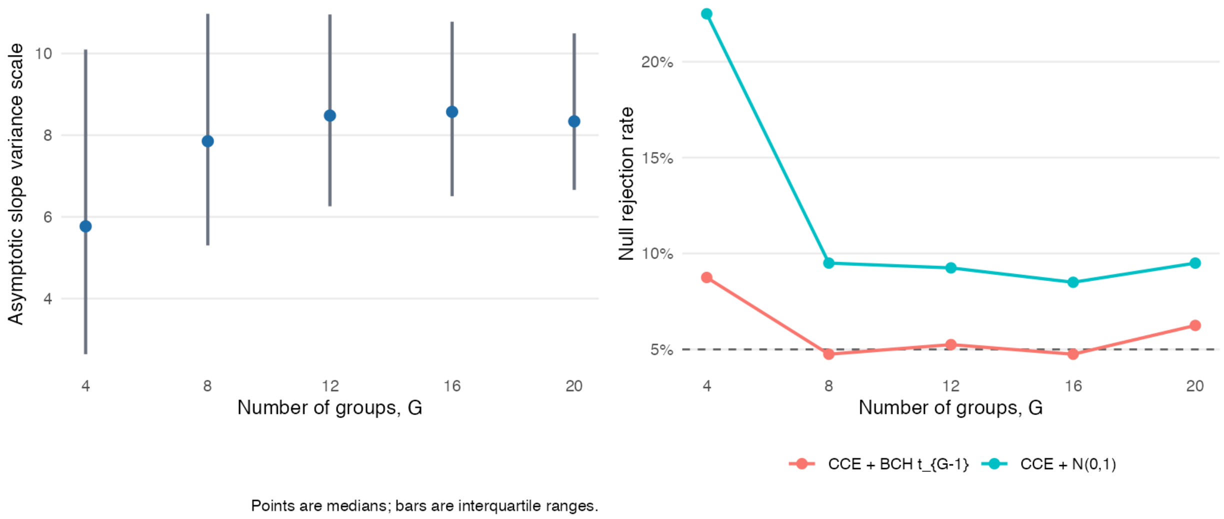

Figure 3: Monte Carlo trade-off across group counts. The design fixes $K = 60$, $\gamma = 0.6$, and uses balanced contiguous blocks with $G = 4, 8, 12, 16, 20$. Left: median and interquartile range of the asymptotic-scale CCE slope variance. Right: empirical rejection rates under the true null.

Figure 3 anticipates the empirical application more directly. Increasing G gives more degrees of freedom for the reference distribution, but it also changes the group scores entering the covariance matrix. The trade-off is therefore not monotone. In this design, the fixed-G BCH test is close to nominal size for several G's, while the normal CCE test overrejects, especially when G is small. The lesson for the HiClimR application is that G is not a tuning parameter for precision. It is a maintained claim about the spatial scale at which score dependence has been absorbed within groups. Larger G can look more precise, but it is credible only if the resulting groups still leave weak cross-group score dependence.

Empirical Application: Climate and Conflict

The application follows the data design in Burke et al. (2024). Their paper studies whether local wealth or national income moderates the effect of temperature shocks on civil conflict in Africa. It uses a yearly $1^\circ \times 1^\circ$ grid-cell panel and estimates conflict models with cell and year fixed effects. For our purpose, the key feature is the spatially disaggregated weather-conflict panel. It gives a real setting in which the identifying variation is weather-based but the appropriate inference units are ambiguous.

We use the monthly weather-conflict layer behind that application. A cell that is hotter than usual in a given month is closer to a natural experiment than a comparison of hot and cold places. We therefore use deviations from each cell's monthly climatology and absorb cell and year-month fixed effects. This sharpens the inference problem. Climate deviations may be exogenous, but they are not spatially independent. Heat anomalies often cover many neighboring cells. Conflict can also co-move within regions.

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Why This Application Tests the Grouping Problem

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Burke et al. (2024) use African grid cells of roughly $1^\circ \times 1^\circ$. This scale is useful for identifying local exposure, but it does not solve inference. Grid-cell clusters are numerous but implausibly independent. Country clusters are meaningful but may not match weather systems. Supranational macroregions are closer to geographers' regional classifications, but still impose boundaries rather than learning them from climate dynamics. The question is therefore: what is the relevant spatial unit of climate-conflict variation?

This setting is informative for group construction for three reasons. First, temperature anomalies follow a physical process. Their spatial co-movement can be measured before any conflict regression. Second, conflict is observed locally, at grid-cell resolution. Third, fixed effects remove permanent cell differences and common monthly shocks. The remaining issue is the covariance structure of the regression scores.

HiClimR offers one answer. It groups cells with similar temperature-anomaly dynamics. It does not group cells with similar conflict histories. This matters because the grouping rule is tied to the treatment process, not the outcome. It avoids choosing clusters to strengthen or weaken the estimated conflict effect.

Table 1: Why the Burke et al. weather-conflict setting is suitable for BCH group diagnostics

Feature of the setting	Econometric advantage	Remaining problem
Weather deviations from local climatology	Supports a natural-experiment interpretation after fixed effects.	Weather shocks remain spatially correlated.
Grid-cell conflict outcomes	Preserves local variation in violence and exposure.	Neighboring cells may not provide independent score information.
Multiple possible spatial scales	Allows comparison of coarse and fine dependence groups.	The number of effective groups is not observed directly.
Exogenous climate co-movement	Permits pre-outcome construction of climate regions.	Climate homogeneity is not identical to score independence.

Our goal is not to replicate the moderation estimates in Burke et al. (2024). We stress-test the inference layer of the weather-conflict design. If the association is precise only under fine groups, the conclusion should be cautious. If it remains precise under coarse groups, the evidence is stronger.

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Data

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The empirical database is the derived weather-conflict panel stored with the project replication files. It was obtained from the replication package for Burke et al. (2024). The cleaning script crops Berkeley Earth monthly temperature data to Africa, assigns $1° \times 1°$ cells to countries using Natural Earth polygons, spatially joins UCDP GED conflict events to cells, expands events to the months they cover, and creates monthly conflict counts and indicators. We retain the public weather-conflict fields and exclude the Atlas wealth and WDI income variables used for the paper's moderation analysis.

The resulting object is a monthly panel of 2,695 African grid cells from 1989 to 2019. Each cell is matched to country identifiers, coordinates, temperature, monthly climatology, and conflict events. The main conflict variable is

$$Y_{it} = \mathbf{1}\{\text{at least one conflict event occurs in cell } i \text{ during month } t\},$$

equal to one if at least one conflict event occurs in cell i and month t. The climate variable is the monthly temperature anomaly

$$A_{it} = temp_{it} - climatology_{im(t)},$$

where $m(t)$ is calendar month. The BCH application uses 1,002,540 cell-month observations. Mean conflict incidence is 2.916%. Mean temperature is $24.738°C$. Mean anomaly is $0.817°C$.

For comparability with the broader project, we also report the annualized panel. This is not the estimation frequency for the HiClimR-BCH exercise. It gives a compact summary of scale. A cell-year is coded as conflicted if any monthly conflict event is observed.

Table 2: Annual panel coverage and main variables

Statistic	Value
Cell-year observations	83,545
Grid cells	2,695
Countries/territories	51
First year	1989
Last year	2019
Annual conflict incidence	0.103
Mean annual temperature	24.738
Mean annual temperature anomaly	0.817
Mean annual conflict events	0.790

Table 3: Descriptive statistics for the annual grid-cell panel

Variable	N	Mean	SD	Min	Median	Max
Conflict events	83,545	0.790	7.559	0.000	0.000	1,365.000
Conflict indicator	83,545	0.103	0.304	0.000	0.000	1.000
Conflict months	83,545	0.350	1.478	0.000	0.000	12.000
Temperature	83,545	24.738	3.433	11.089	25.033	31.763
Temperature anomaly	83,545	0.817	0.430	-0.965	0.835	2.810
Climatology	83,545	23.921	3.409	11.108	24.255	29.972

The tables establish scale, but not the time structure of identifying variation. Figure 4 therefore compares annual aggregation with the monthly variation used in the BCH application.

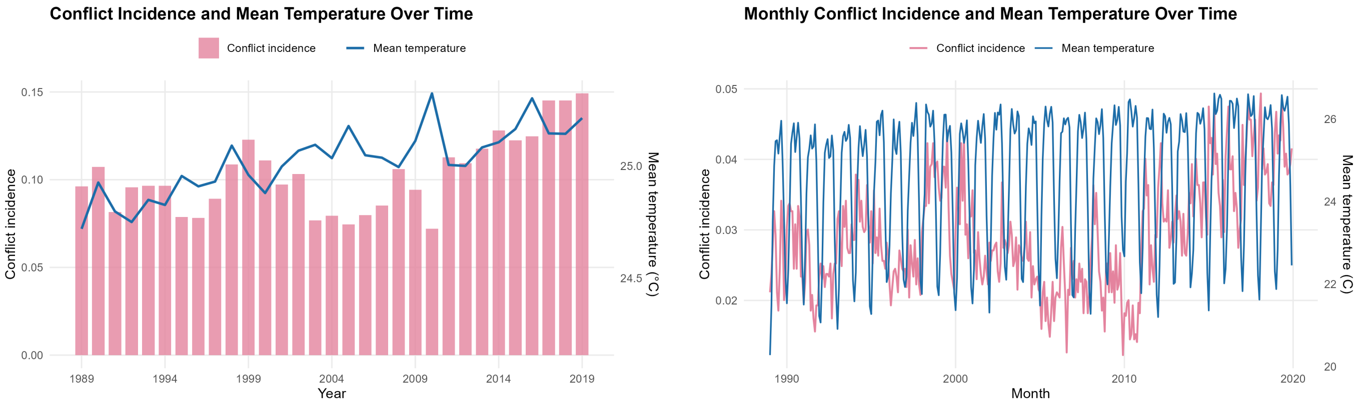
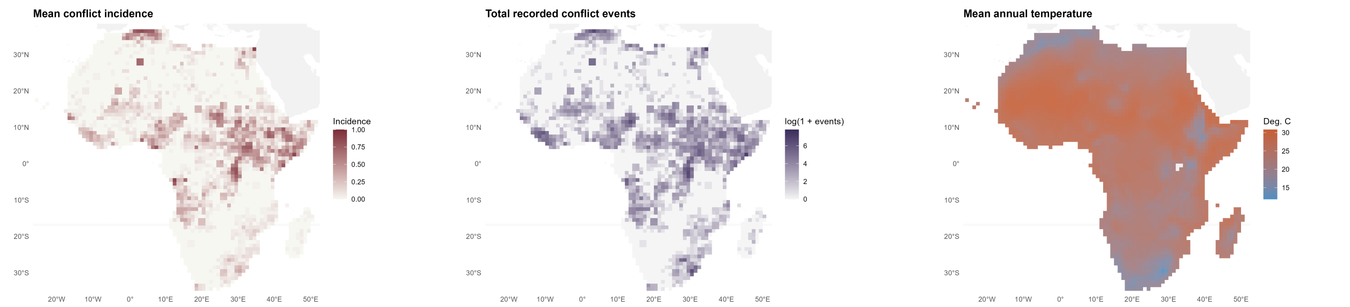


Figure 4: Time-series descriptives from the project copy. The annual view shows slow co-movement; the monthly view preserves the short-run variation used in the BCH application.

Figure 4 motivates the monthly design. Annual data show broad co-movement but compress the shocks that identify β. Monthly data preserve short-run anomalies and provide many observations within each candidate group. The remaining question is spatial: are those observations independent enough across cells? Figure 5 maps the relevant spatial structure.



(a) Conflict incidence (b) Conflict events (c) Mean temperature

Figure 5: Spatial descriptives for the African grid-cell panel. Conflict and temperature both have visible spatial structure. This motivates spatial inference rather than treating cell-months as independent.

Figure 5 shows that neither conflict nor temperature is spatially featureless. This is not evidence that hot places mechanically have more conflict; cell fixed effects remove permanent level differences. The figure matters because nearby cells may share weather systems and unobserved shocks. The monthly panel is therefore useful for BCH, but only with a defensible spatial grouping rule.

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Empirical Specification

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The baseline linear probability model is

$$Y_{it} = \beta A_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where α_i are grid-cell fixed effects and δ_t are year-month fixed effects. The coefficient is identified by within-cell deviations from normal monthly temperature, compared across cells in the same month. The design removes permanent cell differences and aggregate monthly shocks. The coefficient is causal only if residual temperature anomalies are as-good-as random conditional on these fixed effects.

For OLS, the relevant BCH group contribution is the score

$$S_g(\widehat{\beta}) = X_g' \widehat{e}_g,$$

not the outcome alone and not temperature alone. Clustering on temperature is therefore not a substitute for BCH assumptions. It is a way to define large, physically interpretable groups before computing clustered inference.

The broader project establishes the sign and scale of the weather-conflict association. This section asks a narrower question. Does the conclusion survive when independent information is a small number of climate regions rather than thousands of cells?

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HiClimR Climate Groups

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HiClimR is a hierarchical climate regionalization algorithm. We form a $2{,}695 \times 372$ matrix. Rows are cells; columns are months. Entry (i, t) is A_{it}. HiClimR detrends and standardizes each row, computes correlation distances, and builds a clustering tree. Cutting the tree at $G = k$ gives G climate groups.

This addresses the grouping problem directly. Administrative borders are convenient but not physically privileged. Country groups can be too coarse for local violence and too political for weather shocks. Grid-cell clustering is usually too fine for spatial dependence. HiClimR is an intermediate choice: regions follow climate co-movement at the outcome's spatial resolution, without using conflict outcomes.

After the data section shows spatial structure, Figure 6 starts with the coarsest climate grouping. It asks whether the panel can be summarized by a few large regions, as fixed-G inference requires.

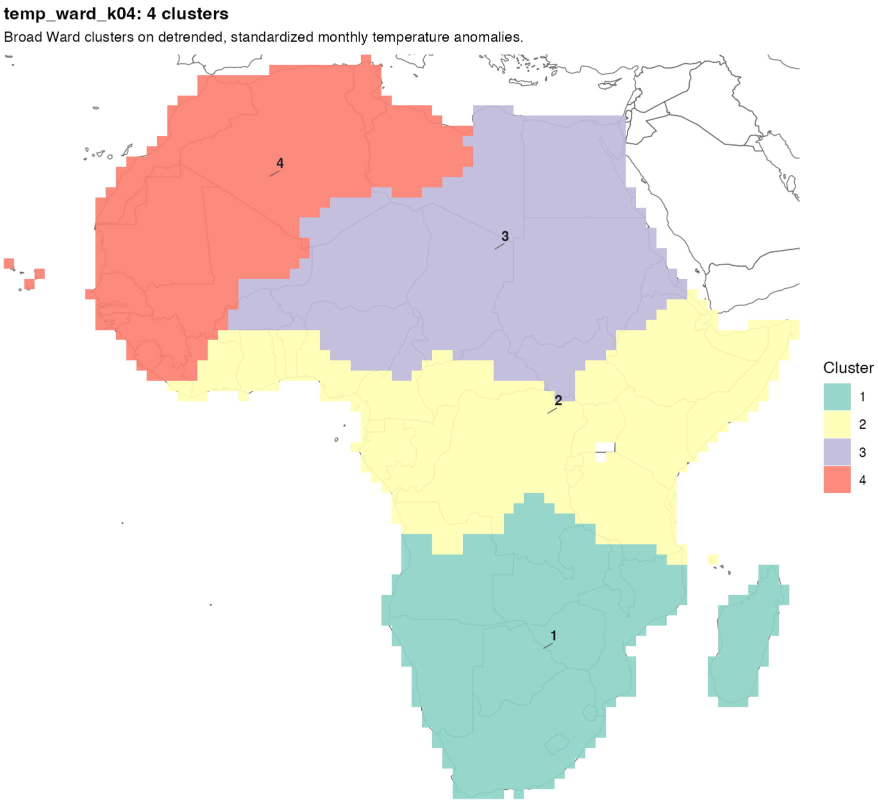

Figure 6: Preferred coarse HiClimR partition. The $G = 4$ Ward partition gives large, nearly contiguous climate groups, which is closest to the BCH requirement of a fixed small number of large groups.

Figure 6 is a benchmark, not proof that $G = 4$ is correct. It shows that climate groups can be large, spatially coherent, and chosen before observing conflict outcomes. This makes the later $G = 4$ inference conservative; finer partitions must be justified by diagnostics, not only by smaller standard errors.

Table 4 reports the candidate partitions. The $G = 4$ partition is most defensible for BCH because all groups are large. The smallest contains 618 cells. Finer partitions raise within-cluster temperature homogeneity, but some groups become small and unbalanced. In fixed-G inference, larger G is not automatically better.

Table 4: HiClimR partitions used as candidate BCH groups

Proposal	G	Cell size min/med/max	Intra corr.	Inter corr.	Method
Broad Ward climate partition	4	618/670/738	0.683	0.239	Ward
Regional-linkage climate partition	8	66/369/636	0.763	0.213	Regional
Contiguity-weighted Ward partition	12	66/196/396	0.816	0.210	Ward + contig.
PCA-filtered Ward partition	16	90/168/279	0.849	0.229	Ward + PCA
Hybrid Ward-regional partition	20	14/124/472	0.855	0.209	Hybrid

Table 4 reports intra- and inter-cluster temperature-anomaly correlations. Higher intra-cluster correlation means more similar anomaly dynamics within groups. Lower inter-cluster correlation means group mean anomaly series are more separated. These are climate diagnostics, not proof of BCH validity. BCH validity concerns the score process. Residual or score diagnostics remain necessary.

The comparison across G is also inferential. $G = 4$ treats Africa as a few large climate regions. $G = 8$ and $G = 12$ keep moderate group sizes. $G = 16$ and $G = 20$ are closer to descriptive regionalization: they improve climate homogeneity but create small groups.

Figure 7 visualizes the trade-off in Table 4: numerical climate homogeneity must still correspond to geographically interpretable groups.

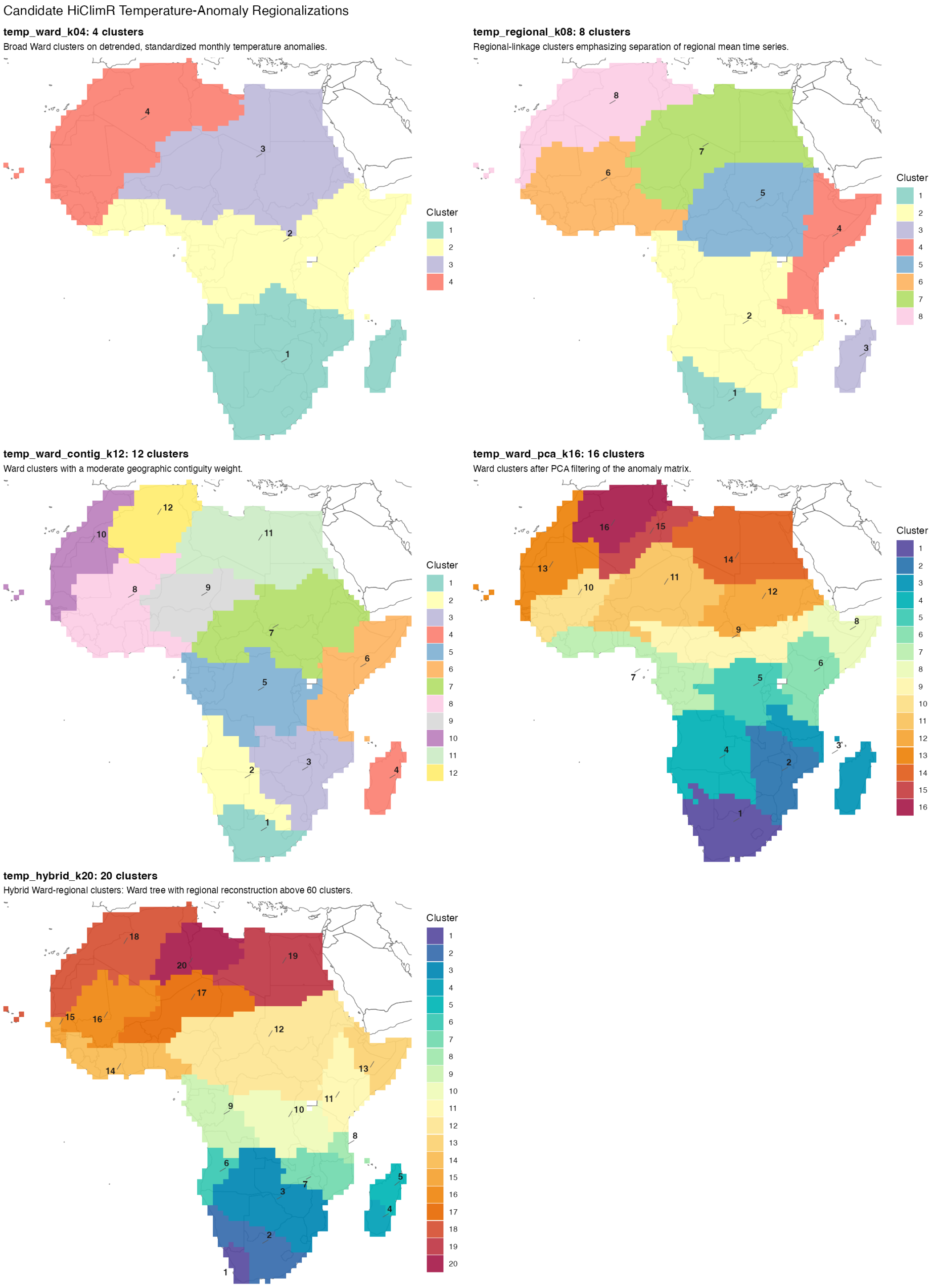

Figure 7: Candidate HiClimR partitions from $G = 4$ to $G = 20$. The figure shows the central trade-off: finer regionalizations add climate detail but produce smaller and less balanced inference groups.

Moving from $G = 4$ to $G = 20$ adds climatic detail, but regions become more fragmented and unequal. BCH does not reward more clusters mechanically. The question is whether groups are large enough for group-score normality and separated enough for weak cross-group dependence. Figure 7 therefore frames the empirical exercise as sensitivity over plausible, but not automatically valid, climate partitions.

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Implementation in R

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The implementation separates estimation from inference. The fixed-effects model is estimated once using monthly conflict incidence as the outcome, monthly temperature anomaly as the treatment, grid-cell fixed effects, and year-month fixed effects. The clustered covariance matrix is then recomputed after assigning each grid cell to a candidate climate region. For each grouping, scalar tests use $G-1$ degrees of freedom, matching the BCH fixed-G reference distribution up to small finite-sample scaling.

The exact software command is reported in the replication code rather than in the main text. This avoids making the essay read like a programming manual while preserving reproducibility. For joint Wald restrictions, BCH derive a scaled F reference; ordinary regression summaries do not automatically impose that scaling.

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Results

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The point estimate does not depend on the inference group because the regression is unchanged. The estimate is

$$\widehat{\beta} = 0.00146.$$

Since Y_{it} is binary, this equals 0.146 percentage points. Relative to the 2.916% baseline, it is about a 5.0% increase for a $1°C$ positive anomaly. Uncertainty depends on G.

Table 5: Temperature anomaly and monthly conflict incidence with HiClimR-group inference

Groups	$\widehat{\beta}$	SE	t	p	95% CI
$G=4$	0.00146	0.00064	2.279	0.107	[-0.00058, 0.00349]
$G=8$	0.00146	0.00046	3.152	0.016	[0.00036, 0.00255]
$G=12$	0.00146	0.00051	2.845	0.016	[0.00033, 0.00258]
$G=16$	0.00146	0.00050	2.919	0.011	[0.00039, 0.00252]
$G=20$	0.00146	0.00050	2.894	0.009	[0.00040, 0.00251]

The preferred interpretation emphasizes $G=4$. It is the coarsest and most balanced partition, so it best matches the BCH intuition. Under this conservative partition, the 5 percent confidence interval includes zero. Finer partitions give smaller p-values, but rely on smaller and less regular groups. The $G=20$ hybrid partition, for example, has a group with only 14 cells. These rows are robustness checks, not the main benchmark.

This is the central empirical result. The point estimate is stable because grouping does not change the regression. Precision changes because grouping changes the information used for inference. The fixed-effects estimate is positive. Under the most conservative grouping, it remains positive but imprecise.

Figure 8 connects the results table back to the simulation. The table reports coefficient inference; the figure shows the variance object behind it. The purpose is not to select the G with the lowest variance, but to show how uncertainty changes when the same score is aggregated into different climate groups.

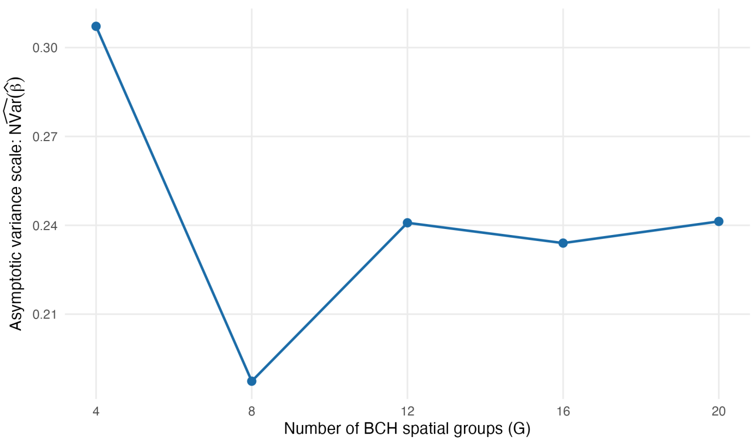

Figure 8: Sensitivity of the asymptotic-scale variance estimate to the number of BCH groups. The non-monotonic pattern is expected: changing G changes both group score aggregation and the finite-G reference distribution.

Figure 8 sharpens the main empirical lesson. The variance estimate is not a smooth decreasing function of G. Changing G changes score aggregation and scalar-test degrees of freedom. The $G=4$ result is cautious because it treats independent information as four large climate regions. Finer partitions make the positive estimate more precise, but that precision is persuasive only if score diagnostics support the implied cross-group independence.

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Score Diagnostics and an Administrative Benchmark

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A defensible BCH application must check whether groups produce approximately independent scores. This is where the data-driven construction can be compared to administrative benchmarks. We use two: five AU-style macroregions and 51 country/territory groups. For the scalar temperature coefficient, let $x_{it} = A_{it}$ after residualizing on fixed effects. The monthly score contribution is approximately

$$s_{it} = x_{it}\widehat{e}_{it}.$$

For each group g and month t, compute

$$S_{gt} = \sum_{i \in g} s_{it}.$$

The diagnostic object is the $G \times G$ correlation matrix of S_{gt} over time. A good grouping has low off-diagonal correlations. No single group should dominate the variance. No obvious neighbor-pair dependence should remain. This is stricter than testing temperature homogeneity.

Table 6: Score-independence diagnostics by grouping rule

Grouping rule	G	Cell size min/med/max	Mean $\lvert\rho\rvert$	Max $\lvert\rho\rvert$	BCH p
HiClimR $G = 4$	4	618/670/738	0.101	0.192	0.107
HiClimR $G = 8$	8	66/369/636	0.088	0.349	0.016
HiClimR $G = 12$	12	66/196/396	0.093	0.353	0.016
HiClimR $G = 16$	16	90/168/279	0.083	0.348	0.011
HiClimR $G = 20$	20	14/124/472	0.079	0.621	0.009
Admin macroregions	5	435/446/822	0.114	0.232	0.076
Countries	51	1/33/219	0.076	0.864	0.012

Table 6 is mixed but useful. Mean absolute score correlations are modest for all rules, so averages alone do not choose the grouping. The maxima and group sizes are more revealing. HiClimR $G = 4$ and administrative macroregions both use a small number of large groups and produce cautious inference. HiClimR $G = 4$ has the lower maximum score correlation, 0.192 versus 0.232, because its boundaries follow climate co-movement rather than inherited regional classifications. Countries provide many degrees of freedom, but include one-cell units and a maximum score correlation of 0.864. They are familiar clusters, not automatically valid BCH groups.

The diagnostic supports a conservative ranking. Data-driven climate groups are not valid because an algorithm produced them. They are relevant because they are chosen from treatment dynamics and then evaluated on regression scores. The central comparison is therefore $G = 4$ HiClimR versus administrative macroregions. Both answer the small-entity correlation problem with few large groups. The climate grouping performs at least as well on score independence and is directly tied to the weather process. Finer HiClimR partitions and country clusters are robustness checks, not automatic improvements.

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Conclusion

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The BCH lesson is that clustered inference is not a mechanical search for many clusters. In spatial panels, a few large, well-shaped groups may be more credible than many fine units. The simulation shows why. With fixed G, the variance estimator remains random, and t_{G-1} critical values are the natural scalar reference.

The application gives a reproducible workflow. HiClimR defines climate regions from temperature-anomaly co-movement. A fixed-effects regression estimates the relationship between monthly temperature anomalies and monthly conflict incidence. Inference is clustered by the chosen climate group. Score diagnostics then ask whether group sums are close to independent. The temperature-conflict association is positive and modest. Its precision depends on the maintained group structure. The most defensible $G = 4$ partition gives cautious evidence, not a decisive rejection of no effect.

The administrative benchmarks clarify what the data-driven approach adds. Macroregions are a credible geographer-style alternative and lead to the same cautious conclusion as HiClimR $G = 4$. Country borders offer many degrees of freedom, but include one-cell units and a very high maximum score correlation. HiClimR does not validate BCH inference by itself. It makes the grouping rule explicit, tied to climate exposure, and testable at the score level.

The final conclusion is therefore conservative. The number of groups matters because it changes both the covariance estimate and the finite-G reference distribution. Group choice matters because formal degrees of freedom are useful only if the corresponding group scores are close to independent. In this application, the positive temperature-conflict association is robust in sign but not in inferential strength. That is exactly the type of distinction fixed-G inference is designed to make visible.

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